

Task 1: Spam Detection using Machine Learning

Objective

Build and evaluate a supervised classification model to classify SMS messages as spam or ham.

Dataset

SMS Spam Collection Dataset with 5572 messages (Ham: 4825, Spam: 747).

Results

Random Forest achieved 97.67% accuracy, 100% precision, 82.67% recall, 90.51% F1, 98.68% ROC-AUC, and 97.61% cross-validation accuracy.

Conclusion

Random Forest was selected as the best model.